

Angus Sun

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EDUCATION

University of Waterloo

Bachelors of Software Engineering (GPA: 3.8/4.0)

Waterloo, ON

Sept. 2025 – Present

EXPERIENCES

WATonomous

Software Engineer

Dec. 2025 – Present

Waterloo, ON

- Scaled WATcloud to support **650+ users** on GPU-backed **Kubernetes** and **SLURM** infrastructure
- Automated cluster and CI provisioning with **Ansible**, reducing duplicated YAML by **50%**
- Implemented **Prometheus** alerts on SMART metrics, detecting boot drive failures **24–72h** faster

Falcon Technologies

Software Engineer Intern

Feb. 2025 – Aug. 2025

Vancouver, BC

- Designed and implemented **RESTful APIs** for a subscription tracker (**Next.js**, **Prisma ORM**)
- Implemented secure authentication using **NextAuth** with **Google OAuth 2.0** and JWT-based sessions
- Dockerized the **MySQL database**, eliminating manual database setup and cutting deploy time by **60%**
- Built a responsive, dashboard UI with **Tailwind CSS**, including auth-gated pages, forms, and tables

PROJECTS

Did I Cook? – Debate Judge Platform 🤖 🌐 | Next.js, Spring Boot, FastAPI, OpenSearch, RAG

- Built a real-time debate platform with structured phases, transcript capture, and end-of-match evaluation
- Engineered a **RAG pipeline** for top-*k* evidence retrieval grounding AI scoring in transcript segments
- Integrated **Google Gemini** judging with strict JSON schema validation and a decoupled **FastAPI** worker, enabling non-blocking evaluation of ~15-minute debate transcripts and preserving real-time debate flow

MatchA Dance – 1st Place (GoOn Hacks Hackathon) 🤖 🌐 | React, Express, PostgreSQL, Supabase

- Built an AI-powered dance battle app with pose-based judging and **80%** accuracy against human ratings
- Developed stateless **Express APIs** and a **PostgreSQL**-backed leaderboard for a **React** and **Vite** frontend
- Configured **Supabase** with Row-Level Security to support user profiles, videos, and full CRUD operations
- Managed **CI/CD pipeline** and deployments on **Vercel** and **Render** (staging/prod) for a 4-person team

Squat Showdown – Best AI Hack (Boost Hacks II Hackathon) 🤖 🌐 | Node.js, WebSockets, MediaPipe

- Designed a state machine for squat detection using **MediaPipe** pose keypoint transitions
- Implemented **WebSocket**-based multiplayer state synchronization with **sub-200ms** end-to-end latency
- Devised pose-based rep counter and scoring logic with consistent agreement against manual rep counts

Skync – Real-time Collaborative Whiteboard 🤖 🌐 | React, WebSockets, Canvas

- Created a **multi-user** whiteboard with real-time drawing, image, and text collaboration via WebSockets
- Developed **telemetry** to measure application-level latency for collaborative strokes at **480–500 ms**
- Built custom canvas logic (selection, resizing, undo/redo) with a modular **React UI** and dynamic toolbars

Nanopeas Website – E-Commerce Site for Non-Profit 🌐 | JavaScript, PayPal API

- Architected a full-stack **e-commerce platform** for a Vancouver nonprofit, peaking at **3.5K** daily users
- Integrated **PayPal API** with automated fee computation and transactional order-processing backend

SKILLS

Languages: JavaScript, TypeScript, Python, Java, SQL, C/C++, HTML/CSS

Technologies: React, Next.js, Node.js, Express, FastAPI, Spring Boot, PostgreSQL, MySQL, Prisma ORM, Supabase, Docker, Kubernetes, Ansible, Prometheus, OpenSearch, Git, Vercel, Render